

Hoja de instrucciones RV32IM

Registros	
x0	zero Siempre contiene el valor 0
x1	ra Dirección de retorno
x2	sp Puntero de pila
x3	gp Puntero global
x4	tp Puntero de hilo
x5-x7	t0-t2 Datos temporales
x8	s0/fp Registro preservado / Puntero <i>frame</i>
x9	s1 Registro preservado
x10-x11	a0-a1 Parámetros entrada/salida
x12-x17	a2-a7 Parámetros de entrada
x18-x27	s2-s11 Registros preservados
x28-x31	t3-t6 Datos temporales
Instrucciones de carga/almacenamiento	
lb rd, imm12, rs1	I $rd \leftarrow {}_8 \text{MEM}[rs1 + \text{sign}(\text{imm12})]$
sb rs2, imm12, rs1	S $\text{MEM}[rs1 + \text{sign}(\text{imm12})] \leftarrow {}_8 rs2$
lh rd, imm12, rs1	I $rd \leftarrow {}_{16} \text{MEM}[rs1 + \text{sign}(\text{imm12})]$
sh rs2, imm12, rs1	S $\text{MEM}[rs1 + \text{sign}(\text{imm12})] \leftarrow {}_{16} rs2$
lw rd, imm12, rs1	I $rd \leftarrow {}_{32} \text{MEM}[rs1 + \text{sign}(\text{imm12})]$
sw rs2, imm12, rs1	S $\text{MEM}[rs1 + \text{sign}(\text{imm12})] \leftarrow {}_{32} rs2$
Instrucciones de salto	
jal rd, imm20	U $rd \leftarrow pc + 4, pc \leftarrow pc + \text{imm20}$
jalr rd, imm12, rs1	I $rd \leftarrow pc + 4, pc \leftarrow rs1 + \text{imm12}$
beq rs1, rs2, imm12	S $\text{if } rs1 = rs2 - pc \leftarrow pc + \text{imm12}$
bne rs1, rs2, imm12	S $\text{if } rs1 \neq rs2 - pc \leftarrow pc + \text{imm12}$
bge rs1, rs2, imm12	S $\text{if } rs1 \geq rs2 - pc \leftarrow pc + \text{imm12}$
blt rs1, rs2, imm12	S $\text{if } rs1 < rs2 - pc \leftarrow pc + \text{imm12}$
bgeu rs1, rs2, imm12	S $\text{if } rs1 \geq_u rs2 - pc \leftarrow pc + \text{imm12}$
bltu rs1, rs2, imm12	S $\text{if } rs1 <_u rs2 - pc \leftarrow pc + \text{imm12}$

Formatos de instrucción

R	func7 – 31..25	rs2 – 24..20	rs1 – 19..15	func3 – 14..12	rd – 11..7	opcode – 6..0
I	imm12 – 31..20		rs1 – 19..15	func3 – 14..12	rd – 11..7	opcode – 6..0
U	imm20 – 31..12					opcode – 6..0
S	imm12(4) – 31..25	rs2 – 24..20	rs1 – 19..15	func3 – 14..12	imm12(2) – 11..7	opcode – 6..0
C	CSR – 31..20					opcode – 6..0

Instrucciones aritméticas	
add rd, rs1, rs2	R $rd \leftarrow rs1 + rs2$
sub rd, rs1, rs2	R $rd \leftarrow rs1 - rs2$
mul rd, rs1, rs2	R $rd \leftarrow rs1 \times rs2$
div rd, rs1, rs2	R $rd \leftarrow rs1 / rs2$
rem rd, rs1, rs2	R $rd \leftarrow rs1 \vee rs2$
addi rd, rs1, imm12	I $rs \leftarrow rs1 + \text{sign}(\text{imm12})$
lui rd, imm20	U $rd \leftarrow \text{sign}(\text{imm20} < 12)$
auipc rd, imm20	U $rd \leftarrow pc + \text{sign}(\text{imm20} < 12)$

Instrucciones lógicas	
and rd, rs1, rs2	R $rd \leftarrow rs1 \& rs2$
or rd, rs1, rs2	R $rd \leftarrow rs1 rs2$
xor rd, rs1, rs2	R $rd \leftarrow rs1 \wedge rs2$
andi rd, rs1, imm12	I $rd \leftarrow rs1 \& \text{sign}(\text{imm12})$
ori rd, rs1, imm12	I $rd \leftarrow rs1 \text{sign}(\text{imm12})$
xori rd, rs1, imm12	I $rd \leftarrow rs1 \wedge \text{sign}(\text{imm12})$

Instrucciones desplazamiento de bits	
sll rd, rs1, rs2	R $rd \leftarrow rs1 << rs2$
srl rd, rs1, rs2	R $rd \leftarrow rs1 >> rs2$
sra rd, rs1, rs2	R $rd \leftarrow rs1 >>_a rs2$
slll rd, rs1, imm	R $rd \leftarrow rs1 << \text{imm}$
srrl rd, rs1, imm	R $rd \leftarrow rs1 >> \text{imm}$
sra! rd, rs1, imm	R $rd \leftarrow rs1 >>_a \text{imm}$

Instrucciones de comparación	
slt rd, rs1, rs2	R $rd \leftarrow rs1 < rs2 ? 1 : 0$
slti rd, rs1, imm12	R $rd \leftarrow rs1 < \text{sign}(\text{imm12}) ? 1 : 0$
slti rd, rs1, imm12	I $rd \leftarrow rs1 < \text{sign}(\text{imm12}) ? 1 : 0$
sltiu rd, rs1, imm12	I $rd \leftarrow rs1 <_u \text{sign}(\text{imm12}) ? 1 : 0$

Otras instrucciones	
ecall	- Invoca <i>sycall</i> indicado en a7
ebreak	- Punto de parada para depurador
rdcycle rd	C $rd \leftarrow$ Número de ciclos
rdtime rd	C $rd \leftarrow$ Ticks transcurridos
rdinstret rd	C $rd \leftarrow$ Instrucciones retiradas

Pseudoinstrucciones	
li rd, imm12	addi rd, zero, imm12
li rd, imm	lui rd, imm[31:12] addi rd, rd, imm[11:0]
la rd, etq	auipc rd, etq[31:12] addi rd, rd, etq[11:0]
mv rd, rs	addi rd, rs, 0
not rd, rs	xori rd, rs, -1
neg rd, rs	sub rd, zero, rs
j imm20	jal zero, imm20

bgt rs1, rs2, imm12	blt rs2, rs1, imm12
bgtu rs1, rs2, imm12	bltu rs2, rs1, imm12
bte rs1, rs2, imm12	bge rs2, rs1, imm12
beqz rs, imm12	beq rs, zero, imm12
bnez rs, imm12	bne rs, zero, imm12
bgez rs, imm12	bge rs, zero, imm12
bgtz rs, imm12	blt zero, rs, imm12
blez rs, imm12	bge zero, rs, imm12
call imm12	jalr ra, ra, imm12
call imm	auipc ra, imm[31:12] jalr ra, ra, imm[11:0]
ret	jalr zero, 0, ra
nop	addi zero, zero, 0

